

PROPOSAL

In-House Spare Parts Portal: Database, Platform and Hosting

Development of a RUF Diamond owned parts database and online catalog, hosted on Canadian infrastructure, covering all three equipment brands, beginning with a single-model pilot.

Prepared by: Deepshika Ghale

Prepared for: RUF Diamond Management

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1. Purpose

This document sets out a plan to develop a spare parts database and online catalog owned and operated by RUF Diamond to replace the current third-party platform. It describes what will be built, how it will be hosted, how the work will be sequenced, where costs arise, and what approvals are required to begin.

The work begins with a single equipment model as a pilot. Nothing is migrated, and nothing is switched off until the pilot is complete and reviewed.

2. Current State

RUF Diamond's online spare parts catalog runs on Intelli Catalog, a third-party platform hosted by the vendor at rufdiamond.intellicatalogue.com, with a brand-specific portal at fattruck.intellicatalogue.com. Neither is part of the main RUF Diamond website.

Item	Current situation
Platform	Intelli Catalog, a vendor-hosted third-party electronic parts catalog system
Database	Owned and structured by the vendor. No direct query access, no integration path, no control of the schema.
Hosting	Vendor-managed. Physical location, provider, jurisdiction, uptime commitments, and backup arrangements are not documented on our side.
Coverage	6 Fat Truck models (2.8 Pickup, 2.8C, 2.8 Wagon, 8x8 Hauler, 8x8 Wagon, 2.4P) and 2 Batch Plant models. IronHorse is not represented.
Content	Exploded diagrams, hotspot-linked parts lists, quantities, CAD pricing, serial-number breakpoints, remarks
Search	Model, VIN or serial number, figure (image), and part number or description
Ordering	Cart and purchase-order functions exist but are restricted. Customers select parts and send screenshots to their purchasing team, or email parts@rufdiamond.com .
Data updates	Routed through the vendor rather than made directly by our team.

Four limitations follow from this arrangement: the parts data cannot be queried, integrated, or exported by us; updates cannot be made on our own schedule; the hosting location and its commitments are undocumented; and the IronHorse line is absent from the catalog.

3. Objectives

- **Ownership of the database.** Parts data held in a schema RUF Diamond controls, on infrastructure RUF Diamond controls, queryable and exportable at any time.
- **Direct updates.** Part numbers, descriptions, quantities, pricing, remarks, diagrams, and serial-number breakpoints are edited directly by our team, with changes live to customers the same day.

- **Current information for customers.** Customers see up-to-date data when searching for parts and placing online spare parts orders.
- **Canadian hosting.** Parts data, pricing, and customer account information held in Canada, under Canadian jurisdiction.
- **Three brands.** Fat Truck, Batch Plant, and IronHorse, with brand separation for users.
- **Separation from the main website.** The portal runs on its own subdomain and its own infrastructure.
- **Integration capability.** Ownership of the database allows later connection to our ERP, automated pricing updates, and online ordering.

4. What Will Be Built

4.1 Components

Component	Function
Parts database	The system of record. Brands, models, serial-number variants, systems and assemblies, figures, parts, positions on each diagram, quantities, pricing, and remarks.
Administration back office	Internal tool. Add and edit parts and pricing, upload diagrams, place hotspots on figures, manage models and serial-number breakpoints, manage customer accounts, view the change log.
Customer catalog	Customer-facing portal. Search by model, serial number, figure image, or part number and description. Hotspots on exploded diagrams link to parts. Zoom, print, email, and part selection.
Ordering	Later stage. Cart and purchase-order submission for approved customer accounts, with per-customer rules where ordering runs through a purchasing department.

4.2 Database structure

The data model reflects how the equipment is structured:

- **Products:** Fat Truck, Batch Plant, IronHorse.
- **Model,** and beneath it, serial-number variants, for example, 2.8C with serial number up to 9928C210502 and 2.8C with serial number 9928C210503 and up.
- **System or assembly:** cabin, drive system, electric, engine, filters, frame, fuel system, hydraulic, tire and wheel, and so on.
- **Figure:** an exploded diagram image belonging to a system. Each reference number on the drawing is stored as a hotspot mapped to a part.
- **Part:** part number, description, attributes, and remarks. Stored once and referenced by every figure and model in which it appears.
- **Position:** joins a part to a figure with its reference number and quantity, allowing the same part to appear at different quantities on different machines.

- Price: held per part with currency and effective date, retaining pricing history.
- Customer accounts, permissions, and an audit log recording every change and who made it.

PostgreSQL is proposed as the database. It carries no licence fee, is widely supported, and is straightforward to back up, replicate, and export. Diagram images are held in object storage alongside it.

Pricing can be fed from the ERP rather than entered twice, and parts data can be exported to a spreadsheet on demand.

4.3 Hosting

The portal runs on its own subdomain, for example, parts.rufdiamond.com, on infrastructure separate from the main RUF Diamond website.

The proposed host is ThinkOn (thinkon.com), a Canadian-owned and Canadian-operated cloud provider headquartered in Toronto. ThinkOn runs a sovereign cloud from Canadian data centres, supported by Canadian-resident staff. It is Canada's first VMware Sovereign Cloud Partner and an approved supplier under Shared Services Canada's cloud framework agreements. It has no foreign parent company, so data held with it is not subject to foreign disclosure legislation such as the US CLOUD Act.

Alternatives, in descending order of preference:

- **Another Canadian-owned sovereign provider** with equivalent certifications and Canadian-resident operations staff.
- **A Canadian region of a major cloud platform** (AWS Canada Central in Montreal, Microsoft Azure Canada Central in Toronto, or Canada East in Quebec City, or Google Cloud in Montreal or Toronto). This satisfies residency but not sovereignty, and carries egress charges.

The environment will meet the following specifications:

Requirement	Specification
Residency and sovereignty	Database, application, and diagram images held in a Canadian data centre operated by a Canadian-owned provider, subject only to Canadian law.
Privacy compliance	Customer account data is handled in line with PIPEDA and Quebec Law 25, where applicable.
Availability	99.9% monthly uptime target, backed by the provider's service level agreement.
Security	HTTPS/TLS on all pages, role-based access control, multi-factor authentication on administrator accounts, database encrypted at rest.
Backups and recovery	Daily automated database backups are retained for at least 30 days, with a tested restore procedure and defined recovery time and recovery point objectives.
Data ownership	The database is a RUF Diamond property. Full export at any time, in standard formats, at no cost.

4.4 Data migration

Before development begins, we establish in writing what can be extracted from Intelli Catalog: parts lists with reference numbers, quantities, pricing, remarks, and the exploded diagram images. The current catalog screens include an Export control, and the contract will be reviewed for data return provisions.

Where structured export is unavailable, the data will be reconstructed from RUF Diamond engineering source documents, which we hold independently of the vendor. This is slower and forms the largest single unknown in the plan.

The existing portal continues to serve customers throughout. Nothing is switched off until the replacement is complete and reviewed.

4.5 Ordering

Once the data is in place, cart and purchase-order submission is enabled for approved customer accounts, replacing the current process in which customers screenshot their part selections and forward them to a purchasing team. Per-customer rules apply, since some customers, including Agnico Eagle, require ordering to run through their own purchasing department. Order submission can later be connected to the ERP. This work is sequenced after the data work.

5. Pilot

That said, our initial plan would be to start with a single equipment model as a pilot project to test how everything works before expanding further.

The database, administration back office, and customer catalog are built, and one model is loaded into them. That model runs live alongside the existing Intelli Catalog portal, with a selected group of customers using it for 60 to 90 days. No data is migrated, no system is switched off, and the current portal continues to serve every other model.

The proposed pilot model is an IronHorse model, since IronHorse is not present in the existing catalog. The pilot therefore adds a machine that is not covered today rather than duplicating one that is, and no model is maintained in two systems at once.

The pilot is complete when the following four conditions are met:

- Our team loads and maintains the pilot model entirely through the administration back office, with no external dependency.
- Changes to parts, pricing, or diagrams are live to customers within one business day.
- Customers locate parts for the pilot model using model, serial-number, figure, and part-number search, at least as easily as on the existing portal.
- Hosting, residency, security, and backup arrangements are documented, and a restore from backup has been tested successfully.

The pilot is then reviewed, and a costed plan is brought back to management for migrating the Fat Truck and Batch Plant catalogs and retiring the vendor platform.

6. Cost

No dollar figures are given below. The build has not been scoped and quoted, and ThinkOn sells through channel partners rather than a published private-sector price list, so any figure stated now would be speculative. Firm numbers are obtained by written quotation before commitment. What follows is where cost arises and where it does not.

6.1 Cost items

Item	Notes
Build of the portal and back office	Principal investment. Quoted by a development partner or resourced internally, against a written specification. Ordering functionality is quoted separately as a later stage.
Canadian hosting	Recurring. Quoted through a ThinkOn partner. ThinkOn's published price list covers Ontario public-sector bodies and does not apply to RUF Diamond.
Data extraction and preparation	Effort to export or reconstruct parts lists, pricing, and diagram images. Dependent on what Intelli Catalog releases. The largest unknown, and the reason for the pilot.
Database software	Nil. PostgreSQL carries no licence fee.
Subdomain	Nil. A DNS record on a domain already owned by RUF Diamond.
SSL/TLS certificate	Nil. Issued through Let's Encrypt and renewed automatically.
Ongoing maintenance	Recurring. Security updates, backup monitoring, and support. Internal, or a retainer with the development partner.

6.2 Hosting cost basis

The hosting specification in Section 4.3 requires managed cloud infrastructure with guaranteed resources, a production database, daily backups with tested recovery, an uptime commitment, SOC 2 Type II or ISO 27001 certification, and support. It is not comparable to shared website hosting of the kind sold by GoDaddy and similar providers, which places a static site on a shared server without those commitments and which, being US-owned, would not meet the sovereignty requirement.